Monday 24 April 2023
14:00-15:30 BST
Duration: 1 hour 30 minutes
Additional time: 30 minutes
Timed exam – fixed start time

DEGREES OF MSc, MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

TEXT AS DATA M
COMPSCI 5096

(Answer all 3 questions)

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1. Question on Tokenisation. (Total marks: 20)

(a) What is the difference between a *word* and a *token*? Why do text processing systems usually operate over tokens instead of words? [2]

Example Answer:

Words are sequences of letters that represent an individual unit of language, while tokens are a sequence of characters that convey some meaning. Unlike words, tokens can consist of partial words (e.g., don't → do n't), punctuation (e.g., $), numbers, and whitespace.

Text processing systems often operate over tokens because (1) they provide a more granular unit of meaning than individual words, and (2) non-word units (e.g., symbols, whitespace, etc.) can convey meaning in text, not just words.

(b) Why do tokenisers differ across languages? Provide an example where a tokeniser for English may not be suitable for another language. [2]

Example Answer:

Conventions of what constitutes a meaningful unit differ across languages. A whitespace-oriented English tokeniser would be ineffective on Chinese text, spaces are not present. Or, in German, compound nouns are prevalent, which would often need to be split up.

(c) Build a byte pair tokenisation vocabulary of size 8 for the text: "mississippi is pie". Show all your steps. [8]

Example Answer:

Step 1:

Text: [m i s s i s s i p p i] [i s] [p i e]

Vocabulary: m i s p e (length: 5)

Step 2:

Most common pair: i s (3)

Text: [m is s is s i p p i] [is] [p i e]

Vocabulary: m i s p e is (length: 6)

Step 3:

Most common pair: is s (2)

Text: [m iss iss i p p i] [is] [p i e]

Vocabulary: m i s p e is iss (length: 7)

Step 4:

Most common pair: p i (2)

Text: [m iss iss i p pi] [is] [pi e]
Vocabulary: m i s p e is iss pi (length: 8)
(Done)

(d) Name two advantages of byte pair tokenisation over traditional rule-based tokenisation, and two advantages of rule-based tokenisation over byte pair tokenisation. [4]

Example Answer:
byte pair over rule-based:

- Can be applied to any suitably large text, without the need for a fluent human in the language.
- Identifies potentially useful patterns from the data that may not be apparent to humans

rule-based over byte pair:

- No need for a large corpus of text from which to learn byte pairs.
- Rules can be developed with linguistic knowledge/intuition, allowing coverage of situations that might be rare or unseen in a corpus

(e) Consider a situation where you are building an automatic writing assistant. It aims to provide grammar suggestions for a human writer as they write. Would you use stemming, lemmatisation, and/or stopword removal in this application? Explain why or why not. [4]

Example Answer:
It is not a good idea to use stemming, lemmatisation, or stopword removal in this application.
Grammatical errors often involve incorrect inflection (i.e., endings) of words; stemming would remove these, making it impossible to identify these errors.
Even though lemmatisation considers the part-of-speech (which you may already predict as part of this application), it also treats words with similar endings as the same tokens, which means that it would not be able to identify errors.
Finally, although stopwords are often unhelpful for classification tasks, they are essential for understanding the structure and grammar of text; removing them means that your system would not able able to identify a variety of grammatical errors.

2. **Question on Language Models. (Total marks: 20)**

Consider the bi-gram language models, **Model X** and **Model Y**, defined as follows:

Model	X	Y	Model	X	Y	Model	X	Y	Model	X	Y
$P(a \mid \langle S \rangle)$	0.1	0.3	$P(a \mid a)$	0.1	0.0	$P(a \mid b)$	0.3	0.3	$P(a \mid c)$	0.1	0.0
$P(b \mid \langle S \rangle)$	0.5	0.3	$P(b \mid a)$	0.3	0.1	$P(b \mid b)$	0.4	0.6	$P(b \mid c)$	0.6	0.4
$P(c \mid \langle S \rangle)$	0.4	0.4	$P(c \mid a)$	0.1	0.8	$P(c \mid b)$	0.2	0.0	$P(c \mid c)$	0.2	0.4
$P(\langle E \rangle \mid \langle S \rangle)$	0.0	0.0	$P(\langle E \rangle \mid a)$	0.5	0.1	$P(\langle E \rangle \mid b)$	0.1	0.1	$P(\langle E \rangle \mid c)$	0.1	0.2

a, b, and c are the tokens for a particular language, and $\langle S \rangle$ and $\langle E \rangle$ indicate the start-of-sequence and end-of-sequence tokens, respectively.

(a) For **Model X** and **Model Y** without smoothing, calculate the perplexity of the sequence "$\langle S \rangle$ a b b a $\langle E \rangle$". Which model better represents the sequence? Show your work. [6]

Example Answer:

Model X

$ce = -\frac{\log(0.1)+\log(0.3)+\log(0.4)+\log(0.3)+\log(0.5)}{5} = -\frac{-3.32-1.73-1.32-1.73-1}{5} = 1.824$

$perplexity = 2^{ce} = 3.539$

Model Y

$ce = -\frac{\log(0.3)+\log(0.1)+\log(0.6)+\log(0.3)+\log(0.1)}{5} = 2.171$

$perplexity = 2^{ce} = 4.503$

Model X is more likely to represent the sequence than **Model Y** because it has a lower perplexity (3.5 vs 4.5).

(b) Using **Model X** without smoothing, apply both greedy generation and beam search (with 2 beams), given the prefix: "$\langle S \rangle$ a c". The beam search should extend each beam using the two most likely tokens. Show your work, and provide the probability of each generated sequence. [8]

Example Answer:

Greedy Generation:

Step 1: b ($p = 0.6$)

Step 2: b b ($p = 0.6 * 0.4 = 0.24$)

Step n: b b b ... ($p = 0.6 * 0.4^{n-1}$)

Final Generated Sequence: b b b ... (indefinitely, $p_n = 0.6 * 0.4^{n-1}$)

Beam Search:

Step 1:

Beam 1: b ($p = 0.6$)

Beam 2: c ($p = 0.2$)

Step 2:
Beam 1.1: b b ($p = 0.6 * 0.4 = 0.24$)
Beam 1.2: b a ($p = 0.6 * 0.3 = 0.18$)
Beam 2.1: c b ($p = 0.2 * 0.6 = 0.2 = 0.12$) – pruned
Beam 2.2: c c ($p = 0.2 * 0.2 = 0.04$) – pruned
Step 3:
Beam 1.1.1: b b b ($p = 0.24 * 0.4 = 0.096$)
Beam 1.1.2: b b a ($p = 0.24 * 0.3 = 0.072$) – pruned
Beam 1.2.1: b a $\langle E \rangle$ ($p = 0.18 * 0.5 = 0.09$)
Beam 1.2.2: b a b ($p = 0.18 * 0.3 = 0.054$) – pruned
Step 4: The student either needs to explain that there is no chance for Beam 1.1.1 to achieve a higher probability than 1.2.1 in subsequent steps, or show it.
Final Generated Sequence: b a $\langle E \rangle$ ($p = 0.09$)

(c) Why is smoothing often applied to language models? Which values of **Model X** and **Model Y** would most benefit from smoothing? What are the negative effects of too much smoothing?

[3]

Example Answer:

Smoothing eliminates the effects of sequences with zero (or extremely low) probability, which can occur due to sparsity in the training data. In reality, no sequence of words has no probability.

The following values would benefit from smoothing for **Model Y**: $P(a|a)$, $P(c|b)$, $P(a|c)$. Smoothing would affect $P(\langle E \rangle | \langle S \rangle)$ for both models, but this is not valuable, since it represents a completely empty sequence.

Too much smoothing dilutes the real probabilities and makes unusual words more likely to appear which can create spurious or nonsensical outputs.

(d) For an n-gram language model, what are the problems of a very small or very large value of n? How would you select the best n to use for a particular problem?

[3]

Example Answer:

Too small an n means that limited context is taken into account when modelling [1 mark]. Too large an n means that too much context is taken in and the likelihood of an n-gram matching another n-gram severely reduces or the computational burden of a large n-gram model becomes too severe.

To decide an optimal value of n, an experiment should be set up with some held-back test data for the specific problem.

CONTINUED OVERLEAF

3. Question on the Factuality of Language Model Responses. (Total marks: 20)

Recall that neural language models can generate responses to prompts (such as questions), but that they often make factual errors. We prompt GPT3 with various questions related to this course and provide the response. **Critique each response, and where applicable, correct any factual errors present.**

(a) **Question:** Provide and explain the two probabilities that language models commonly measure. [3]

GPT3 Response:

1. Perplexity: Perplexity is a measure of how well a language model predicts the next word in a sentence. It is calculated by taking the average log probability of the words in the sentence. A lower perplexity score indicates that the language model is more accurate.
2. Coverage: Coverage is a measure of how many words in a corpus (a collection of written or spoken language) are correctly predicted by the language model. A higher coverage score indicates that the language model is able to accurately predict more words from the corpus.

Example Answer: Even though the definitions of perplexity and coverage are correct, the response does not answer the question.

Language models measure:

1) $P(w_1, w_2, ..., w_n)$, The probability of a whole sequence of text w.

2) $P(w_n | w_1, w_2, ..., w_{n-1})$, conditional probability of the next word in a sequence w.

(b) **Question:** Compare and contrast TF and IDF. [2]

GPT3 Response:

TF (term frequency) and IDF (inverse document frequency) are two methods of determining the importance or relevance of a given word or phrase within a document.
TF is a measure of how often a given word appears in a document. It is calculated by dividing the number of times the word appears in a document by the total number of words in the document. The higher the TF value, the more important the word is in the document.
IDF is a measure of how important a given word is in a collection of documents. It is calculated by taking the logarithm of the number of documents in the collection divided by the number of documents in the collection that contain the word. The higher the IDF value, the more important the word is in the collection.
In summary, TF measures the importance of a word within a single document, while IDF measures the importance of a word across a collection of documents.

Example Answer: The answer is mostly correct and provides a reasonable comparison of TF and IDF.

TF is usually used as a raw count (with potential smoothing), rather than as a ratio of the term count and the length of the document.

(c) Question: What role does dimensionality reduction play when building dense word embeddings from sparse word vectors? [4]

GPT3 Response:

Dimensionality reduction can help accelerate the training process of building dense word embeddings from sparse word vectors by reducing the number of dimensions of the input space, making it easier to work with and process. This can reduce the computational complexity of the training process, allowing more efficient and faster training. Additionally, by reducing the dimensions of the input space, the model can more easily discern the contextual information associated with each word, as the noise created by irrelevant dimensions is removed.

Example Answer: The answer is mostly wrong – it does reduce the number of dimensions and makes the vectors easier to process, but it has nothing to do with training and does not help discern contextual information.

Dimensionality reduction instead compresses the information present in sparse vectors while maintaining the positional relationships between them, saving on memory and processing requirements when using them. Further, the dense vectors are easier to work with because they are all the same size.

(d) Question: Why are Hidden Markov Models used for Part-of-Speech Tagging? [3]

GPT3 Response:

Hidden Markov Models are used for Part-of-Speech Tagging because they are probabilistic models that can capture the probability of transitioning from one part-of-speech tag to another. This helps the model better estimate which tag is most likely given a certain context. Additionally, the Hidden Markov Model allows for the use of Viterbi algorithm, which further helps increase the accuracy of the tagging process.

Example Answer: The answer is correct, but leaves out some important information that's important for understanding the answer. For many languages, the syntax imposes rules that make some sequences of POS tags more likely than others, for instance, nouns frequently follow determiners, and verbs frequently do not. By modelling these probabilities, the HMM can produce higher-quality POS tags than if it does not consider the POS of the preceding token(s).

(e) Question: Give an example of a nested entity mention, and explain what the nested entity mentions are. [3]

GPT3 Response:

Example: ”The CEO of Apple, Tim Cook, announced...”
Explanation: The nested entity mention in this sentence is ”Tim Cook”, which is comprised of two separate entities - ”CEO” and ”Apple”. The combination of these two entities forms the nested entity mention ”Tim Cook”.

Example Answer: The answer includes an example of a nested entity mention, but does not correctly identify it. Here, Tim Cook is not nested within another entity mention. Instead, “The CEO of Apple” is itself an entity (a person), while “Apple” (nested within it) is also an entity. (CEO may also be considered an entity, nested within the larger entity mention.)

(f) Question: Explain how GPT3 is able to provide knowledge (such as the current CEO of Apple is Tim Cook) in its answers. What are two challenges of a language model encoding knowledge? [5]

GPT3 Response:

GPT3 is able to provide knowledge by using a large-scale language model that is trained on a massive amount of data. The language model is able to capture patterns in the data and use them to generate answers to questions. For example, it can look at data about Apple and its executives and use that to answer questions about the current CEO of Apple. Two challenges of a language model encoding knowledge are accuracy and generalizability. Accuracy refers to how accurately the language model can answer questions. Generalizability refers to how well the language model can answer questions about topics that it has not seen before.

Example Answer:

The answer is on the right track, but not a very complete answer.

The explanation of how GPT3 is able to provides knowledge doesn’t go into sufficient detail on how the knowledge itself is encoded; it should really mention that knowledge is encoded in model parameters. The idea that the model ”looks at data about Apple” is simply wrong.

The response on accuracy is fine, but could go into more detail about why the accuracy is an issue. In terms of Generalizability, the response really should mention that GPT3 and others tend to hallucinate content it is unable to generalise to.

Students can also get marks for mentioning the following:

- The problem of updating knowledge (e.g. when Tim Cook is replaced as CEO)
- A lack of explainability of how this knowledge is encoded

- The inability to provide a source for the knowledge
- The ethical problems that we are unable to check what knowledge is encoded, and its likely biases

END OF QUESTION PAPER